MANAGEMENT REVIEW

Management without theatre.



OPERATIONS

THE KPI TRAP: WHEN THE NUMBERS IMPROVE BUT THE WORK GETS WORSE

Six chapters on targets, distorted incentives, hidden trade-offs, statistical variation and the scorecard that protects the customer.

2.3% can miss a 1% business target without crossing a hypothetical statistical upper control limit. A signal is not a verdict.

P-CHART MODEL, INVENTED EXAMPLE

INSIDE

COVER STORY
The green score, the broken promise
PAGE 05

THE MODEL
Target versus control limit
PAGE 19

FIELD TOOL
The Honest KPI Card
PAGE 28

EDITORIAL · VOLUME 6

A green score can hide a bad day

Measurement is necessary. The problem begins when the number becomes easier to defend than the purpose it was supposed to serve.

Operations managers do not have the luxury of managing only by instinct. A departure must be on time or late. A load must be complete or incomplete. A room must be ready or not. Every shift creates facts that deserve a record. Yet those records become dangerous when an indicator designed to reveal the work is treated as a complete judgment on the people and the service. The author has worked with last-mile measures such as Delay, Damage, Incomplete and Loading Time, and with reporting and handovers in hospitality. These contexts explain why a KPI can be both useful and incomplete. This sixth volume does not claim that any named employer manipulated a target, hid a defect or punished staff on the basis of a misleading dashboard. Its operational scenes are constructed illustrations, clearly marked wherever an event or measurement is invented. We begin with a deceptively positive score. Campbell and Goodhart explain why high-pressure targets can weaken the measures intended to guide management. Holmström and Milgrom show the incentives problem when some tasks are measurable and others less visible. Bevan and Hood provide a documented public-service case of gaming risk. Statistical process control distinguishes a management target from evidence of a changed process. The final chapters put the measures back into a balanced, auditable decision.

Stiven Janaqi EDITOR · STIVEN CATALYST

HOW EVERY CHAPTER IS BUILT

1 **The living scene**
A shift, a ramp, a desk at 03:00: one piece of work followed as it actually happens, with the hand-off where it breaks. Scenes are constructed from real categories of work and say so.

2 **The conceptual and scientific breakdown**
The mechanism behind the scene, named with the model that explains it, and the research that tests it: the study, its population, its year and its limits, before it is applied anywhere else.

3 **The operating protocol for action**
What a working team can try on Monday: fields, questions, a cadence and a count, each with an owner, a next step and a return of information to the people who do the work.

THE READING CONTRACT

Field
Real categories of operational work, no invented personal incidents.

Evidence
Research setting, date, method and limits stated together.

Action
A KPI decision must survive a customer-and-quality check.

CONTENTS

The chapters, in three layers

CHAPTER I The green score, the broken promise 05

1 SCENE		2 BREAKDOWN		3 PROTOCOL	
The green score, the broken promise	05	The measure is not the mission	06	Read the green before you celebrate	07

CHAPTER II When a measure becomes a target 08

1 SCENE		2 BREAKDOWN		3 PROTOCOL	
The target changes the room	08	Goodhart is a warning, not a law of fate	09	Keep the target from eating the measure	12
		What gets measured gets effort	10		
		When measurement becomes a game	11		

CHAPTER III The hidden cost of one KPI 13

1 SCENE		2 BREAKDOWN		3 PROTOCOL	
The fast process and the extra work	13	A balanced set is not a crowded dashboard	14	Put a guardrail beside the goal	16
		One measure improves. Another changes.	15		

CHAPTER IV The red number and the wrong blame 17

1 SCENE		2 BREAKDOWN		3 PROTOCOL	
The day the dashboard turns red	17	A business target is not a control limit	18	Investigate the red without inventing a culprit	20
		The red point inside the expected range	19		

CHAPTER V The scorecard that tells the truth 21

1 SCENE		2 BREAKDOWN		3 PROTOCOL	
One average, two different days	21	How both groups improve while the total falls	22	Define the KPI before defining success	24
		A scorecard with a conscience	23		

CHAPTER VI Beyond the target 25

1 SCENE		2 BREAKDOWN		3 PROTOCOL	
The meeting that asks one more question	25	Five questions before a verdict	26	Beyond the target: an operating review	27

ALSO IN THIS VOLUME

04 RESEARCH NOTE — What a KPI study is allowed to say

28 FIELD TOOL — The Honest KPI Card

29 CLOSING ESSAY — Responsible for the purpose

30 METHOD — Studies, models and methods

31 SOURCES — Sources and method

RESEARCH NOTE

What a KPI study is allowed to say

A correlation between a target and a behaviour is not proof that every target causes manipulation.

This volume combines several kinds of evidence: theoretical papers on how control pressure or unequal measurement can alter behaviour, observational public-service research on documented risks in one institutional setting, statistical quality-control guidance on telling routine variation from unusual signals, and improvement guidance on practical families of measures. They produce no universal percentage for KPI gaming, establish no falsified result by anyone in the author's work, and do not show that a three-metric dashboard always prevents harmful incentives. They justify a disciplined set of questions for local verification.

01

Observed

Publisher-identified finding, population, date and method.

02

Modelled

Mathematical example with assumptions and reproducible inputs.

03

Proposed

Editorial control, owner and test, awaiting local evidence.

A green score can be both mathematically correct and operationally inadequate if the definition excludes a relevant customer consequence. Likewise, a red score can be a legitimate warning without proving misconduct or an unusual process shift. The reader should ask what the numerator and denominator mean, who is excluded, when events are counted and what happens outside the visible boundary.

OUR READING

This is an evidence-informed operational treatise, not a statistical study of the author's workplaces.

Sources: Donald T. Campbell, Evaluation and Program Planning, 1979 (via Dartmouth occasional paper (1976), reprinted in the Journal of MultiDisciplinary Evaluation (2011)); Bengt Holmström & Paul Milgrom, Journal of Law, Economics, & Organization 7 (special issue), 1991; Gwyn Bevan & Christopher Hood, Public Administration, 2006; NIST & SEMATECH, 2026; Institute for Healthcare Improvement, 2026

Campbell (1979), Holmström and Milgrom (1991), Bevan and Hood (2006), the NIST/SEMATECH e-Handbook and the IHI; all numerical operational cases are invented.

CHAPTER I · THE GREEN SCORE, THE BROKEN PROMISE

The green score, the broken promise

The dashboard is satisfied. The receiving person is not.

HYPOTHETICAL LAST-MILE SCENE, NO DOCUMENTED INCIDENT

At a constructed loading station, a manager sees that departures are occurring faster than last week. The Loading Time indicator is green. The team has earned the result shown on its screen: the recorded interval from the defined start to the defined end is shorter. A delivery issue appears later. One part of an order has not reached the person who expected it. The departure statistic remains unchanged because the defect belongs to a different measure. Nothing in this scene proves that loading speed caused the missing item. The shortfall could have begun in preparation, recording, sorting, loading or a later handover. The managerial temptation is to tell the easy story: one measure improved and another person made a mistake. That is precisely the story that must be tested rather than announced.

The author's documented last-mile work included Loading Time, Incomplete, missing bags and mixed trolleys as categories of operational attention. The event, its sequence and its outcome here are fictional. We use them only to ask what a metric can certify, and what it cannot.

OUR READING

A metric can be accurate within its boundary and still tell too little about the customer's experience.

Editorial composite of real operational categories; conceptual background in the IHI's outcome, process and balancing measures. The essay named below reads a machine's report the same way.

More in the essay The report a machine wrote

CHAPTER I · THE GREEN SCORE, THE BROKEN PROMISE

The measure is not the mission

Every indicator has a definition; every service has a larger purpose. A loading-time measure concerns one interval in a process. An incomplete-delivery measure concerns a different failure mode, and its denominator may count orders, bags or another unit. Both can be meaningful without being interchangeable. A percentage with an ambiguous denominator is not yet decision-grade evidence.

The Institute for Healthcare Improvement distinguishes outcome, process and balancing measures. Outcome asks whether the service produces the intended result. Process asks whether the critical steps are working. Balancing asks whether the intervention creates new trouble elsewhere. This is a useful design lens outside healthcare, but the actual measures and acceptance criteria must be defined within the local operation.

01	**Process**	Loading interval, scan-completion rate or a defined on-time handover.
02	**Outcome**	Complete, correct and timely service for the recipient.
03	**Balancing**	Missing items, rework, safety deviations or new downstream waiting.

It is tempting to merge these numbers into a single score. But a weighted average can conceal a severe outcome unless the weights, veto conditions and customer priorities are explicit. Some failures should never be traded away against faster throughput. The definition of an acceptable result comes before the dashboard design.

OUR READING

Ask what changed for the customer before declaring that an internal target has been won.

Source: Institute for Healthcare Improvement, 2026

IHI, Establishing Measures; the logistics mapping is editorial, not validated at an identified site.

CHAPTER I · THE GREEN SCORE, THE BROKEN PROMISE

Read the green before you celebrate

A five-step review that tests a favourable KPI against the end-to-end service.

1. **Name the indicator boundary**
   Operational start, end, exclusions, time window, numerator, denominator and data source.

2. **Check the underlying work**
   Are the events counted as good complete and usable downstream? Sample anonymised cases where permitted.

3. **Pair the customer outcome**
   Read the service outcome and a balancing measure next to the improved process KPI.

4. **Test plausible alternatives**
   Case mix, volume, cut-off rules, staffing and exceptional days, before assigning a cause.

5. **Document the decision**
   What was confirmed, what is still unknown, who will verify it and when the team will hear back.

When Loading Time improves, also read completeness, on-time service and safety records. Two numbers moving together prove no cause; check that they share time windows and units, since a same-day process number and a next-day customer result may need aligning.

OUR READING

A green number should begin a short verification, not end the conversation.

Source: Institute for Healthcare Improvement, 2026

Editorial operating protocol after the IHI's measurement guidance; no claim of a formal employer-wide implementation.

CHAPTER II · WHEN A MEASURE BECOMES A TARGET

The target changes the room

A measurement programme can alter the very activity it is intended to observe.

CONSTRUCTED TEAM MEETING, NOT A RECORDED DISCUSSION

In a fictional operations review, a manager announces that the only figure shown at the weekly meeting will be one rate of punctual departure. People naturally begin to focus their attention on the recorded departure event. Questions once asked about preparation accuracy, delayed information and recovery after a late start receive less time. The score remains useful, but the meeting has become narrower. No dishonest action is necessary for this shift to happen. People respond to what receives attention, praise and consequences. Staff may do exactly the work the formal measure encourages while leaving less measured responsibilities unattended. The risk is not a defect in the arithmetic; it is a defect in the design of the incentive system.

This is not an allegation about any named employer or person. The point is to introduce a documented theoretical concern: when we convert a diagnostic indicator into a strong control target, its relationship to the work can change.

OUR READING

The stronger the consequence attached to one indicator, the more carefully its blind spots must be examined.

Constructed scenario; theory in Goodhart (1975), Campbell (1979) and Holmström and Milgrom (1991). The essay named below asks which KPIs a team trusts.

More in the essay KPIs the team trusts

CHAPTER II · WHEN A MEASURE BECOMES A TARGET

Goodhart is a warning, not a law of fate

The familiar slogan is a later generalisation of an earlier observation about statistical regularities.

Charles Goodhart observed in monetary-policy analysis that a regularity used for control can cease to behave as it did before control. A popular paraphrase, associated with Marilyn Strathern's discussion of audit culture, says that a measure can cease to be a good measure when it becomes a target. These are related formulations, not two quotations from the same original text. Neither means that every target must fail. Their use in this volume is to challenge the assumption that a measure remains neutral after rewards and penalties are attached. Donald Campbell's 1979 treatment of planned social change also reminds evaluators that observed indicators may fail to reveal whether an intervention really caused the intended improvement. In complex systems, changing conditions and other interventions may move the same measure. A good trend can be real, an artefact of changed definitions or an effect of other influences.

01

Proxy

The number represents only some qualities of the outcome.

02

Pressure

Visibility and consequences change incentives and attention.

03

Validity

Recheck whether the proxy still describes the outcome.

OUR READING

The problem is not using a target. It is forgetting that the target changes the context in which the measure is produced.

Sources: Charles Goodhart, 1975 (via Bevan & Hood (2006), quoting the 1984 reprint); Marilyn Strathern, European Review, 1997; Donald T. Campbell, Evaluation and Program Planning, 1979 (via Dartmouth occasional paper (1976), reprinted in the Journal of MultiDisciplinary Evaluation (2011))

Goodhart (1975) as quoted by Bevan and Hood (2006); Strathern (1997) for the paraphrase; Campbell (1979). The three cards are editorial.

CHAPTER II · WHEN A MEASURE BECOMES A TARGET

What gets measured gets effort

Multitask incentive theory explains why narrow scores can crowd out responsibilities that are difficult to quantify. Holmström and Milgrom's 1991 multitask principal-agent analysis studies how the design of rewards influences the allocation of effort across duties. In a job with several responsibilities, strong incentives tied to only the measurable ones can divert attention from the others. This is a theoretical mechanism, not an empirical measurement of the author's colleagues.

A dispatcher does more than watch one dashboard: they coordinate, verify exceptions and communicate. A night auditor does more than generate a report: open work must be conveyed accurately to the next shift. A wholesale account manager does more than accept orders: availability and promises must remain credible. In each setting, some valuable activity is harder to capture in a clean daily percentage.

01	**Last-mile: on-time departures are easy to count**	Downstream completeness is easy to miss.
02	**Hotel handover: the report was sent**	Whether the next shift can act is easy to miss.
03	**Wholesale: orders accepted**	Promise fulfilment is easy to miss.

The rows are an editorial illustration of how task definitions can differ, not a claim that any of the three organisations paid incentives on these measures. A reasonable scorecard should include the harder-to-see responsibilities, through sampling or qualitative verification where numerical proxies are inadequate.

OUR READING

When effort has several purposes, a single rewarded number rarely describes all of them.

Source: Bengt Holmström & Paul Milgrom, Journal of Law, Economics, & Organization 7 (special issue), 1991

Holmström and Milgrom, Journal of Law, Economics, & Organization 7 (special issue), 1991; the work examples are interpretive.

CHAPTER II · WHEN A MEASURE BECOMES A TARGET

When measurement becomes a game

Published evidence from public services illustrates risks without proving identical behaviour in private operations.

Gwyn Bevan and Christopher Hood studied target-based governance in the English public healthcare system and described reported success alongside measurement problems and gaming risk. Their 2006 paper examines a particular policy environment, with high-stakes public targets and accountability arrangements. Its context matters: hospital services, political governance and public reporting are not the same as a privately operated loading ramp or hotel shift. Sandra van Thiel and Frans Leeuw described the performance paradox in the public sector: growing quantities of performance information may coexist with declining insight into actual performance, tunnel vision or analysis paralysis. Again, this is not evidence that the author has witnessed gaming. It is a reason to examine what the chosen indicator excludes and how the organisation responds to it.

The practical response is not to accuse individuals whenever a score looks unusually good. Investigate the metric definition, any changes in eligibility or timing, the quality of supporting records, the incentives attached to the score and what independent customer outcomes say.

OUR READING

A surprising score is a question for verification, not an accusation against the team.

Sources: Gwyn Bevan & Christopher Hood, Public Administration, 2006; Sandra van Thiel & Frans L. Leeuw, Public Performance & Management Review 25(3), 2002

Bevan and Hood, Public Administration 84(3), 2006; van Thiel and Leeuw, Public Performance & Management Review 25(3), 2002.

CHAPTER II · WHEN A MEASURE BECOMES A TARGET

Keep the target from eating the measure

Five safeguards before an indicator is attached to praise, ranking or consequences.

1. **Keep a diagnostic copy**
   The measure used to learn stays separate from the number used to judge.
2. **Name what the proxy cannot see**
   The responsibilities that get no number, and how they will be sampled.
3. **Add a counter-measure**
   A balancing indicator that would worsen if the target were pursued too narrowly.
4. **Freeze the definition**
   Eligibility, timing and exclusions versioned before the first comparison.
5. **Review the incentive, not only the score**
   Ask periodically what the target has started to reward, and what it has stopped noticing.

An unusually good score deserves the same review as a bad one: the definition, the records behind it, any change in eligibility or timing, and what independent customer outcomes say. None of this presumes misconduct; it keeps the measure worth trusting.

OUR READING

A target is safe to use only while someone still checks what it has stopped measuring.

Sources: Charles Goodhart, 1975 (via Bevan & Hood (2006), quoting the 1984 reprint); Donald T. Campbell, Evaluation and Program Planning, 1979 (via Dartmouth occasional paper (1976), reprinted in the Journal of MultiDisciplinary Evaluation (2011)); Bengt Holmström & Paul Milgrom, Journal of Law, Economics, & Organization 7 (special issue), 1991; Gwyn Bevan & Christopher Hood, Public Administration, 2006

Editorial protocol after Goodhart, Campbell, Holmström and Milgrom and Bevan and Hood; the author's PDF folds these safeguards into its research page.

CHAPTER III · THE HIDDEN COST OF ONE KPI

The fast process and the extra work

Local speed can improve while an adjacent task inherits additional effort.

HYPOTHETICAL SERVICE FLOW, NO MEASURED EMPLOYER RESULT

Imagine a team that reduces the time needed to close an incoming request. The dashboard shows more requests marked as processed before the deadline. A downstream colleague finds that some requests now arrive without information necessary to complete the service. The originating team may be doing the measured part faster, while the receiving team spends more time clarifying incomplete handovers. Neither the handover defect nor its cause is established by this short story. A proper inquiry would compare a sample of records before and after the change, verify whether the same type of requests were counted and distinguish a real process-design problem from a coincidence. The team needs a measure of work usable at the next step, not only the date at which the first step was marked finished.

This scenario applies to orders, shift handovers and support requests in principle. It is deliberately not a factual claim about specific guest complaints, transactions, route adjustments or a documented process intervention.

OUR READING

Faster completion at the first desk is not necessarily faster delivery to the customer.

Hypothetical comparison; conceptual sources the IHI's balancing measures and Kaplan and Norton (1992). The essay named below follows a part of an order that went missing.

More in the essay Where part of the order goes missing

CHAPTER III · THE HIDDEN COST OF ONE KPI

A balanced set is not a crowded dashboard

The point is to protect the purpose of the work with complementary measures, not to measure everything. Kaplan and Norton introduced the Balanced Scorecard in 1992 to broaden attention beyond traditional financial measures, combining perspectives on customers, internal processes and organisational capability. It is a strategic management approach, not a rule that every operational dashboard must reproduce four boxes unchanged. A useful front-line scorecard can be simpler.

The IHI's outcome, process and balancing distinction provides a concrete test. If one team is asked to improve a process indicator, the review should include a result experienced by the customer and a countermeasure that can detect unwanted costs elsewhere. Measurement may also require a qualitative sample, because a tally of completed forms cannot tell whether the receiving colleague understands the open case.

01

Outcome

The result a customer or receiving team actually values.

02

Process

A controllable step plausibly connected to that result.

03

Balancing

What might deteriorate if the process score is pushed.

More measures can also create more noise. Choose a small family with different jobs; remove duplicates; define ownership; decide how often to review each. Any safety-critical exception should be visible in its own right rather than averaged away by a more favourable speed figure.

OUR READING

A balanced scorecard is a discipline of looking at the whole result, not an excuse for multiplying targets.

Sources: Robert S. Kaplan & David P. Norton, Harvard Business Review, 1992; Institute for Healthcare Improvement, 2026

Kaplan and Norton, Harvard Business Review, January–February 1992; IHI, Establishing Measures.

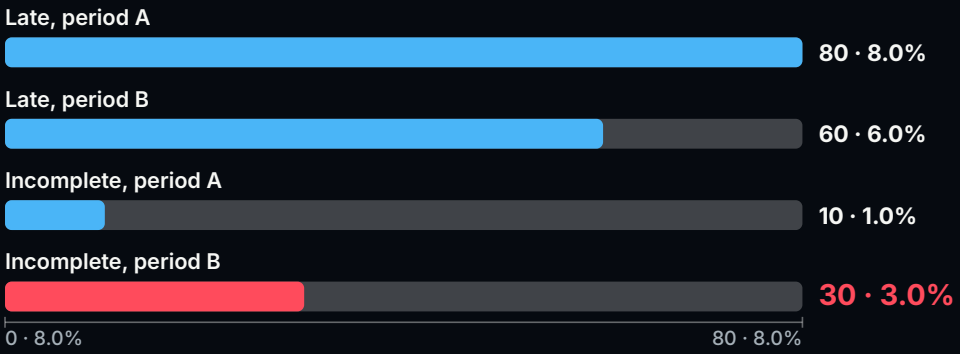
CHAPTER III · THE HIDDEN COST OF ONE KPI

One measure improves. Another changes.

A completely synthetic numerical example shows why two indicators must be examined together: two periods of 1,000 orders each.

SYNTHETIC EXAMPLE, AUTHORED FOR THIS VOLUME; NOT EMPLOYER DATA

FREQUENCY PER 1,000 ORDERS, INVENTED



The late-order rate falls by two percentage points while the incomplete-order rate rises by two points. We have described two facts about the artificial table. We have not demonstrated that faster departures caused incomplete orders, nor decided whether one percentage point has the same value in the two outcomes. Also, a late and an incomplete order may be the same order: the counts cannot simply be added to infer a unique number of affected orders. A genuine analysis would need event definitions, overlap handling, confidence considerations, customer severity, comparable volume and evidence about the process change. Without those, the apparent trade-off is only a prompt for investigation.

OUR READING

Do not add percentages with different meanings, and do not infer a cause from two lines on a chart.

Source: Institute for Healthcare Improvement, 2026

Every value on this page is synthetic, authored for this volume; measure taxonomy after the IHI.

CHAPTER III · THE HIDDEN COST OF ONE KPI

Put a guardrail beside the goal

A practical protocol for reviewing an improvement without relocating its cost.

1. **State the primary goal**
   The customer effect and the operational metric intended to lead toward it.
2. **Name a possible side effect**
   One or two plausible downstream or safety consequences.
3. **Align the measurement scope**
   Match population, period and unit of analysis; record known missingness.
4. **Decide before the pilot**
   Review triggers and non-negotiable stop conditions where policy warrants them.
5. **Check both sides**
   The result, the balancing signal and a small sample of actual work.
6. **Preserve the distinction**
   Worked, partially worked or not yet knowable: record which.

For a hotel handover, a timely report is one thing; whether the next shift has enough actionable information is another. For last-mile loading, departure timing and completeness may deserve paired attention. These are design proposals, not already verified combined KPIs for the author's workplaces.

OUR READING

You do not need a perfect dashboard to avoid celebrating a problem transferred to another team.

Sources: Institute for Healthcare Improvement, 2026; Robert S. Kaplan & David P. Norton, Harvard Business Review, 1992

Editorial operating protocol from the IHI's measure families and Kaplan and Norton's balanced-performance lens.

CHAPTER IV · THE RED NUMBER AND THE WRONG BLAME

The day the dashboard turns red

A worse number may require attention now. It does not always justify changing the process tomorrow.

CONSTRUCTED MANAGER REVIEW, NO CLAIMED EVENT

A manager sees a red daily rate on a performance screen. It is above a target. The instinct is to ask who caused it, and then to call an immediate meeting. But one daily proportion contains several questions that the colour cannot answer: How many opportunities were counted? Has the definition changed? Does the process normally fluctuate? Is this a genuine exceptional event or one observation in an already unstable system? The answer also depends on the nature of the defect. A serious safety incident, a compliance breach or a direct risk to customers demands immediate action according to policy even if no statistical alarm has occurred. For ordinary process improvement, however, reacting to every red point as proof of an assignable cause can create needless changes and more variability.

The author has worked with daily performance discussions, loading deviations and hotel reports, but there is no verified personal case in which a particular red cell caused the precise sequence described here.

OUR READING

A red colour is an invitation to verify. It is not a complete explanation.

Constructed scene; the statistical distinction follows the NIST/SEMATECH e-Handbook. The essay named below asks whether it was a bad day or a problem.

More in the essay A bad day or a problem?

CHAPTER IV · THE RED NUMBER AND THE WRONG BLAME

A business target is not a control limit

Two horizontal lines can look identical on a chart and answer different questions. A business target expresses a desired service level. A statistical control limit is derived from a model of how a process varies when its underlying mechanism is sufficiently stable. A daily value may fail the target while remaining within expected variation; a sudden signal can occur while the value still meets the target.

NIST describes control charts as measurements plotted over time with a centre line and control limits. The limits are chosen according to statistical assumptions and the chart type. For proportions of nonconforming units, a p-chart may be appropriate if the counted opportunities and dependence assumptions support it. A target set by management is neither the centre line nor the upper control limit.

01

Target

What performance we want or require.

02

Control limit

What the process model predicts as routine variation.

03

Safety trigger

What policy requires immediate action regardless of the chart.

Some processes do not justify a Shewhart chart without more work. Varying case mix, serial dependence, changing denominators or an unstable baseline can invalidate a simple formula. Control charts are tools for disciplined investigation, not instant verdicts on individuals.

OUR READING

Missed target, unusual signal and unsafe condition are three separate management questions.

Sources: NIST & SEMATECH, NIST/SEMATECH e-Handbook of Statistical Methods: What are Control Charts? (section 6.3.1), 2026; NIST & SEMATECH, NIST/SEMATECH e-Handbook of Statistical Methods: Proportions Control Charts (section 6.3.3.2), 2026

NIST/SEMATECH e-Handbook, sections 6.3.1 and 6.3.3.2.

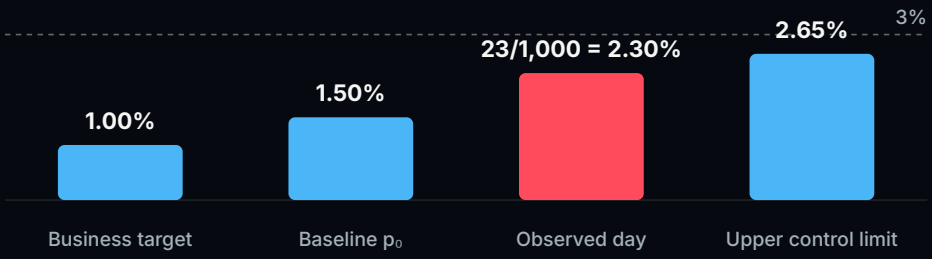
CHAPTER IV · THE RED NUMBER AND THE WRONG BLAME

The red point inside the expected range

An illustrative p-chart calculation with fixed sample size and explicit assumptions: $UCL = p_0 + 3\sqrt{[p_0(1 - p_0)/n]}$, with $p_0 = 0.015$ and $n = 1{,}000$ opportunities per day. The upper 3-sigma limit is about 0.0265 (2.65%); the lower limit is about 0.0035 (0.35%).

SYNTHETIC BASELINE, ASSUMING INDEPENDENT BINOMIAL COUNTS

DAILY SHARE OF INCOMPLETE ORDERS, PER CENT, INVENTED



The day at 2.30% fails the business target but, under this simplified chart, does not cross the upper control limit. That does not certify the day as acceptable: statistical evidence of a changed mechanism is a different question from service adequacy. The calculation assumes a stable historical proportion, independent opportunities and consistent definitions, conditions that real data must be checked for. The UCL is a conventional p-chart calculation, not an employer threshold, and no real baseline is disclosed.

OUR READING

An undesirable rate can be ordinary for a poorly performing process. Routine does not mean good.

Source: NIST & SEMATECH, 2026

Synthetic p-chart example from the NIST proportions-chart formula: $3\sqrt{(0.015 \times 0.985 / 1{,}000)} \approx 0.0115$, so UCL ≈ 2.65% and LCL ≈ 0.35%. No local baseline or test result is claimed.

CHAPTER IV · THE RED NUMBER AND THE WRONG BLAME

Investigate the red without inventing a culprit

A proportionate response to a data signal, a service breach or a safety risk.

1. **Check immediate risk**
   Safety, guest-protection and compliance rules first, whenever a hazard is active.
2. **Validate the entry**
   Source, timestamp, denominator, inclusion rules, pending corrections.
3. **Locate the question**
   Target miss, unusual process signal, recurring customer harm, or several?
4. **Talk to the process owners**
   What conditions were present, from senders and receivers; no blame first.
5. **Test an explanation**
   Comparable periods, case mix and direct observation before changing the routine.
6. **Close the response**
   Verified, unresolved, next action and who receives feedback.

A team should not have to defend itself against a different root-cause story each morning; nor may statistical caution excuse ignoring a dangerous condition or a chronic service failure. The decision fits the risk and the quality of the evidence.

OUR READING

The right response is not always faster. It is better matched to what the signal actually tells us.

Source: NIST & SEMATECH, 2026

Editorial protocol informed by NIST control-chart guidance and common operational escalation principles.

CHAPTER V · THE SCORECARD THAT TELLS THE TRUTH

One average, two different days

When the mix of work changes, an aggregate KPI can tell the opposite story from its components.

PURELY NUMERICAL ILLUSTRATION, NO EMPLOYER DATA

A manager reviews two periods of service. In each category of work, performance improves. The simpler requests are completed more reliably; the complex requests are also handled more reliably. Yet the overall on-time percentage falls. No one has falsified a count. The dashboard has changed because the proportion of difficult work has changed dramatically. The manager who reads only the total may conclude that the team has deteriorated. The colleague who reads the categories may conclude that the process has improved. Both can point to correct arithmetic. The missing information is the composition of the denominator. The next page makes the effect visible with synthetic integer counts that anyone can recalculate.

The editorial lesson extends across last-mile, hotel occupancy patterns and wholesale orders: different kinds of work do not necessarily require the same effort, create the same risk or occur in the same mix each day. A fair analysis must preserve the definition and case mix of each cohort.

OUR READING

An average can be correct and still fail to describe the operational change that matters.

Conceptual mathematical illustration of aggregation effects; no implied observation from the author's workplaces. The essay named below explains why KPIs do not improve in Excel.

More in the essay KPIs do not improve in Excel

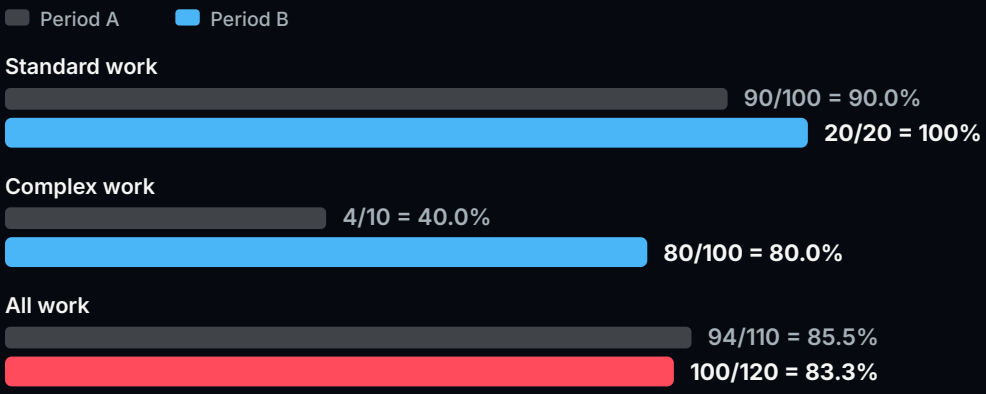
CHAPTER V · THE SCORECARD THAT TELLS THE TRUTH

How both groups improve while the total falls

A recomputable example of an aggregation reversal caused by changed case mix. Success means on-time completion.

SYNTHETIC EXAMPLE, AUTHORED FOR THIS VOLUME; NOT EMPLOYER DATA

ON-TIME SHARE BY CATEGORY AND IN TOTAL, INVENTED COUNTS



Both categories improve, yet the overall on-time share falls from about 85.5% to 83.3%, because the demanding category is a far larger share of period B. The totals are not like-for-like comparisons: the reversal is an arithmetic property of the weights, not a real trend. A metric that supports decisions about people should show whether the mix changed and report volumes; segment where risks differ, with sufficient samples and stable definitions, because an arbitrary split creates noise of its own.

OUR READING

Before comparing two totals, ask whether they contain the same kind of work.

All numbers are invented; the computation was checked from integer numerators and denominators: 94/110 = 85.45%, 100/120 = 83.33%.

CHAPTER V · THE SCORECARD THAT TELLS THE TRUTH

A scorecard with a conscience

The measure family must connect operational action to the customer and make its exclusions visible. A useful scorecard is not a table in which every indicator has equal status. Some establish service adequacy; some show process health; some warn of side effects; some qualify interpretation by showing volume, case mix and coverage. A threshold can prompt review without becoming a statement of guilt.

01	**Outcome**	Complete and timely service; customer-visible quality.
02	**Process**	Departure or handover steps that have a causal rationale.
03	**Balancing**	Incomplete orders, rework, unsafe shortcuts or complaint recurrence.
04	**Context**	Volume, case mix, exception exposure and seasonality.
05	**Data quality**	Coverage, exclusions, delays, corrections and definition changes.

Balanced Scorecard thinking can inform strategy, while IHI measure families help design local improvement. The manager can add a data-quality note: a well-defined rate based on partial records may be precise without being representative. The percentage of records with a valid status is therefore sometimes as important as the reported service rate. Personal identifiers, guest details and confidential operational records should not be placed in a general dashboard; aggregated or demonstrative figures can explain the method without exposing internal information.

OUR READING

The scorecard should make a disputed number easier to investigate, not harder to challenge.

Sources: Robert S. Kaplan & David P. Norton, Harvard Business Review, 1992; Institute for Healthcare Improvement, 2026; NIST & SEMATECH, 2026

Conceptual synthesis after Kaplan and Norton (1992), the IHI's measurement guidance and NIST's data assumptions.

CHAPTER V · THE SCORECARD THAT TELLS THE TRUTH

Define the KPI before defining success

A compact metric dictionary protects decisions from silent changes to the denominator.

1. **Purpose and user**
   Who needs the number, and which decision could it support?
2. **Unit and formula**
   Numerator, denominator, units, event boundary and time window.
3. **Source and freshness**
   The authoritative source, refresh schedule, known latency and ownership.
4. **Eligibility and exceptions**
   Exclusions, cancellations, repeats, late entries and case-mix segmentation.
5. **Companion measures**
   Pair the process metric with a customer result, balancing guardrail and data-quality check.
6. **Govern the changes**
   Version the definition, retain comparable history and announce changes before comparing periods.

Do not apply an invented definition of Delay or Incomplete to a company whose official metric dictionary has different rules. Nor should the meaning of a hotel report be inferred merely from its title. A professional publication may describe general structures, but a real decision requires the authorised local definition.

OUR READING

The most important part of a KPI may be the small print that explains what is missing.

Source: Institute for Healthcare Improvement, 2026

Editorial metric-dictionary protocol, not an existing approved data-governance standard of any employer.

CHAPTER VI · BEYOND THE TARGET

The meeting that asks one more question

A useful performance review asks for the service result after it sees the number.

HYPOTHETICAL MANAGEMENT REVIEW, NO CLAIM OF DIRECT EXPERIENCE

A constructed leadership team arrives at a weekly review. One KPI is better than last week. The slides are ready; the celebration is almost immediate. Instead, the chair asks whether the rate is based on comparable work, whether the count is complete and whether any downstream indicator worsened. The analyst can answer two questions but not the third. A small sample is assigned to the process owner, and the outcome is left open until it can be checked. This is not indecision. It is a proportionate refusal to turn an incomplete statistic into a complete claim. The meeting still recognises what is known, and it still honours the people who improved a measured step. It simply avoids attributing a business result that the evidence has not established.

The fictional meeting describes the practice this volume advocates. It is not a reconstructed conversation with supervisors or colleagues from the author's career.

OUR READING

The strongest dashboard is the one that makes an honest “we do not know yet” possible.

Constructed scenario, inspired by statistical process control, balanced measurement and evidence discipline. The essay named below is about talking to someone who made a mistake.

More in the essay How to talk to someone who made a mistake

CHAPTER VI · BEYOND THE TARGET

Five questions before a verdict

A manager can take accountability for a decision without pretending the metric contains the entire explanation.

01	**Is it defined?**	Can another competent reader reproduce the formula, population and period?
02	**Is it trustworthy?**	Are records complete, current and consistent with the physical or service process?
03	**Is it different?**	Is the change practically meaningful and, where appropriate, an unusual statistical signal?
04	**Is it balanced?**	Did the customer result improve without unacceptable safety, quality or downstream costs?
05	**Is it actionable?**	Does the responsible team have authority to change the condition causing the result?

The sequence matters. Starting with a new target before validating a denominator creates false urgency. Turning a signal into an accusation before comparing the process creates distrust. Adding ten new metrics without removing stale ones creates work but not insight. Every metric review should have an agreed route for corrections: a corrected number should not quietly replace the old one without an audit trail. If the measure is right but the process is poor, the problem belongs to process improvement; if the indicator is neither reliable nor useful, it may need redesign.

OUR READING

A KPI deserves trust because its meaning and evidence can be inspected, not because it is displayed in colour.

Sources: Donald T. Campbell, Evaluation and Program Planning, 1979 (via Dartmouth occasional paper (1976), reprinted in the Journal of MultiDisciplinary Evaluation (2011)); NIST & SEMATECH, 2026; Institute for Healthcare Improvement, 2026

Editorial framework drawn from Campbell (1979), NIST process-control methods and the IHI's measurement guidance.

CHAPTER VI · BEYOND THE TARGET

Beyond the target: an operating review

A proposed four-week pilot, with no assumption that every problem can be resolved in a month.

Week 1	Week 2	Week 3	Week 4
Define.	Validate.	Observe.	Review.

1. **Week 1: choose a real decision**
   One KPI behind a recurring managerial choice, with its official definition.
2. **Week 2: audit a small sample**
   Freshness, coverage, duplicates, timing and a companion service measure.
3. **Week 3: trace discrepancies**
   Observe work where a misleading reading may arise; protect staff privacy.
4. **Week 4: agree the decision rule**
   Owner, escalation conditions, follow-up date, a guardrail against side effects.
5. **After the pilot: compare and explain**
   Findings, unknowns, changes; the definition revised only with documented governance.

Four weeks are an editorial test cadence, not a proven duration or an employer procedure: a safety-critical finding is escalated at once, a structural redesign may take longer. The output is a more defensible choice and a team that knows what the number means.

OUR READING

Measurement serves management only when management remains responsible for judgment.

Sources: Institute for Healthcare Improvement, 2026; NIST & SEMATECH, 2026

Original editorial pilot after IHI and NIST guidance; not a certified or validated intervention.

FIELD TOOL

The Honest KPI Card

One indicator, one decision and an explicit check on what the measure cannot see. Use blank or synthetic data in public examples.

01	**Purpose and decision**	who uses it, and what choice is it meant to support?
02	**Exact definition**	numerator, denominator, time window, exclusions and metric owner
03	**Source and freshness**	authoritative dataset, update lag and known gaps
04	**Signal or target?**	business threshold, control-chart assumptions or urgent risk trigger
05	**Balance and customer impact**	what downstream quality, safety or service measure could worsen?
06	**Action and feedback**	next verification, named owner, checkpoint and report-back

Sources: Institute for Healthcare Improvement, 2026; NIST & SEMATECH, 2026; Robert S. Kaplan & David P. Norton, Harvard Business Review, 1992

Do not add employee, guest or payment identifiers. If a question has no valid answer, record “not measured” or “not yet verified” rather than fabricating a precise result. One page is enough to expose the difference between a number and a decision.

Open the tool: KPI Diagnostic stivencatalyst.com/tools/kpi-diagnostic/

CLOSING ESSAY

Responsible for the purpose

The purpose of a metric is to improve the work, not to replace the work with an attractive description.

A manager needs numbers. Without them, recurring failures can remain anecdotal, improvement becomes difficult to test and the quieter parts of the organisation may never be seen. But measurement also creates a temptation. Once a target is placed on a wall, people begin to talk about it as though it were the purpose of the operation. A green cell offers relief. A red cell invites an explanation. Both feelings arrive before the evidence is necessarily complete. What matters is whether the number still connects to the service it was invented to describe. In the last-mile setting, that means remembering that loading, departure, completeness and customer outcome are linked but not interchangeable. In a hotel, the report and the handover must support work that continues across shifts. In wholesale, an accepted order and a reliable promise are separate accomplishments. The relevant measures differ, but the responsibility to check their meaning is shared.

Campbell reminds us that planned improvement is difficult to evaluate. Goodhart warns that applying pressure can alter the usefulness of a regularity. Holmström and Milgrom explain why disproportionate incentives on measured tasks can distort attention. Bevan and Hood describe risks in a documented public-service setting. IHI and NIST offer tools for building a more balanced and statistically honest review. None of these removes the manager's duty to listen to people, inspect the process and protect the customer. A metric need not be perfect to be useful. It needs a defined scope, a trustworthy source, a companion signal and a person willing to say what is not yet known. When it stops guiding a better decision, its colour becomes a decoration.

OUR READING

A target can be achieved while the purpose of the work is lost. The manager is responsible for noticing the difference.

Editorial conclusion. No manipulation, hidden defect or sanction is attributed to any named employer.

METHOD

Studies, models and methods

An evidence-informed synthesis, not a field experiment. The last-mile and hotel contexts reflect categories of work described by the author; every stated meeting, individual event and numerical operational dataset is explicitly synthetic.

01	**Theory**	Goodhart (1975, via Bevan and Hood), Strathern (1997), Campbell (1979) and Holmström and Milgrom (1991) describe mechanisms, not measured rates.
02	**Observation**	Bevan and Hood (2006) and van Thiel and Leeuw (2002) concern public services; their findings are not transferred as outcomes of logistics or hospitality.
03	**Method**	Kaplan and Norton (1992), the IHI's measure families and NIST's control-chart sections are design guidance; the p-chart is calculated under its stated assumptions.
04	**Arithmetic**	80, 60, 10 and 30 per 1,000; UCL ≈ 2.65% and LCL ≈ 0.35% for p_0 = 0.015 and n = 1,000; 94/110 = 85.5% and 100/120 = 83.3%: all re-checked.

The Honest KPI Card and the four-week cycle are editorial proposals for local testing, not certified interventions. Editorial standard: a theory, an observation, a model and a proposal are named as such, and the sentence stops claiming where the evidence stops.

Sources: Charles Goodhart, 1975 (via Bevan & Hood (2006), quoting the 1984 reprint); Marilyn Strathern, European Review, 1997; Donald T. Campbell, Evaluation and Program Planning, 1979 (via Dartmouth occasional paper (1976), reprinted in the Journal of MultiDisciplinary Evaluation (2011)); Bengt Holmström & Paul Milgrom, Journal of Law, Economics, & Organization 7 (special issue), 1991; Gwyn Bevan & Christopher Hood, Public Administration, 2006; Sandra van Thiel & Frans L. Leeuw, Public Performance & Management Review 25(3), 2002; Robert S. Kaplan & David P. Norton, Harvard Business Review, 1992; Institute for Healthcare Improvement, 2026; NIST & SEMATECH, NIST/SEMATECH e-Handbook of Statistical Methods: What are Control Charts? (section 6.3.1), 2026; NIST & SEMATECH, NIST/SEMATECH e-Handbook of Statistical Methods: Proportions Control Charts (section 6.3.3.2), 2026

Every source is cited at its DOI or page on the next page; the author's PDF gives the p-chart's lower limit as clipping to zero, which the site corrects to about 0.35%.

SOURCES AND METHOD

Where the claims come from

The figures in this volume come from the sources below, each with its year. The scenes are constructed; the protocols are proposals; what is research is named as research.

Assessing the impact of planned social change
Donald T. Campbell, Evaluation and Program Planning, 1979 · via Dartmouth occasional paper (1976), reprinted in the Journal of MultiDisciplinary Evaluation (2011)

Problems of Monetary Management: The U.K. Experience
Charles Goodhart, 1975 · via Bevan & Hood (2006), quoting the 1984 reprint

'Improving ratings': audit in the British University system
Marilyn Strathern, European Review, 1997

Multitask Principal-Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design
Bengt Holmström & Paul Milgrom, Journal of Law, Economics, & Organization 7 (special issue), 1991

The Balanced Scorecard: Measures That Drive Performance
Robert S. Kaplan & David P. Norton, Harvard Business Review, 1992

What's measured is what matters: targets and gaming in the English public health care system
Gwyn Bevan & Christopher Hood, Public Administration, 2006

The Performance Paradox in the Public Sector
Sandra van Thiel & Frans L. Leeuw, Public Performance & Management Review 25(3), 2002

Establishing Measures (Model for Improvement)
Institute for Healthcare Improvement, 2026

NIST/SEMATECH e-Handbook of Statistical Methods: What are Control Charts? (section 6.3.1)
NIST & SEMATECH, 2026

NIST/SEMATECH e-Handbook of Statistical Methods: Proportions Control Charts (section 6.3.3.2)
NIST & SEMATECH, 2026



MANAGEMENT REVIEW

Management without theatre.

Every volume, one question about operations, followed from the floor to the protocol and tested against research.

ALL VOLUMES

stivencatalyst.com/magazine/operational-treatise.html

Operational Treatise · Volume 6 · October 2026 · Stiven Catalyst
Edited by Stiven Janaqi, who answers for it; “the editors” in this publication means him. Its texts, translations, layout and the checking of figures against their sources are done in working sessions with Claude, an AI model by Anthropic.
stivencatalyst.com/how-this-is-made.html